Two counterexamples to a force–energy bound

16 October 2026

The conclusion contradicted here is **Theorem 4(iii), equation (7)**, of Herman, Knorn and Ahlén [1], printed page 15. We give an analytic five-follower counterexample and an exact-arithmetic, computer-assisted counterexample with 128 followers and zero initial state. The latter is an exact polynomial replacement of the numerical witness, with a certified relative violation that rounds to 7.26%.

1 The claim and its hypotheses

Let U have ones on its first superdiagonal and zero elsewhere, and put

$$B = I - U, \qquad K = BB^T, \qquad D_h = (B + h|B|)B^T.$$

Here $|B|$ means entrywise absolute value. We take all masses, stiffness coefficients and damping coefficients equal to one: in the paper's notation $M = A = R = I$. Position coupling is symmetric, $h_\Delta = 0$, and velocity asymmetry is $h_p = h > 0$. With a stationary leader, momentum p and spacing error Δ, the paper's system (4)–(5) becomes

$$\dot p = -D_h p + B\Delta + d, \qquad \dot\Delta = -B^T p, \qquad H(p,\Delta) = \tfrac12(\|p\|^2 + \|\Delta\|^2). \tag{1}$$

All unqualified vector and operator norms are Euclidean. In equilibrium displacement coordinates, $p = \dot x$ and $\Delta = -B^T x$, so equivalently

$$\ddot x + D_h\dot x + Kx = d, \qquad H = \tfrac12(\|\dot x\|^2 + \left\|B^T x\right\|^2). \tag{2}$$

In this specialization, the claimed bound is

$$H(t) \le H(0) + \frac{J}{2\sigma_{\min}(D_h)}, \qquad J = \int_0^\infty \|d(t)\|^2\, dt. \tag{3}$$

The integral is over the whole half-line. Both forces below vanish after their stated terminal time T, so $J = \int_0^T \|d\|^2$.

Hypothesis or convention	Check for both examples
Assumption 2: uniformly positive, bounded m_i, a_i, r_i	All are one, for every index and every flock size.
Assumption 3: $r_i \ge r_{i+1}$	Equality holds.
Symmetric position coupling; $h_p > 0$	$h_\Delta = 0$; $h_p = 1$ in §2 and $h_p = 1/2$ in §3.
Leader and terminal boundary	Leader velocity is zero; the last follower uses only its predecessor, exactly as encoded by B in the source.
Admissible disturbances and initial states	Forces have compact support and lie in L^2. Theorem 4(iii) permits arbitrary initial data; §3 starts at zero.

The problem in the source proof is its equation (8): for nonsymmetric D_h, the relevant quadratic-form lower bound uses $S = (D_h + D_h^T)/2$, not $\sigma_{\min}(D_h)$. Indeed,

$$\dot H = -p^T D_h p + p^T d = -p^T S p + p^T d.$$

Failure of that proof step alone would not disprove (3). The two examples below contradict the integrated inequality itself.

2 An analytic counterexample with five followers

Take $N = 5$, $h = 1$, so $D_1 = 2B^T$. Set

$$p_* = (3, 5, 6, 5, 3)^T, \quad \Delta(0) = 0, \quad p(0) = p_*, \quad d(t) = \begin{cases} p_*/2, & 0 \le t \le T, \\ 0, & t > T, \end{cases} \quad T = \frac{1}{20000}. \tag{4}$$

In particular, no velocity coupling has a negative weight; the rear velocity weight is zero when $h = 1$.

Lemma 1. *For this matrix,* $\sigma_{\min}(D_1) > 5/9$, *hence* $1/(2\sigma_{\min}(D_1)) < 9/10$.

Proof. Let $C = B^{-1}$, the upper triangular matrix of ones. Then $(C^TC)_{ij} = \min(i, j)$ and

$$\operatorname{tr}\left((C^TC)^2\right) = 9 + 28 + 45 + 48 + 25 = 155.$$

Since $D_1^{-1} = C^T/2$, the largest eigenvalue of the positive matrix $D_1^{-1}D_1^{-T}$ satisfies

$$\sigma_{\min}(D_1)^{-4} \le \operatorname{tr}\left((D_1^{-1}D_1^{-T})^2\right) = \frac{155}{16}.$$

Thus $\sigma_{\min}(D_1)^4 \ge 16/155 > (5/9)^4$; the last comparison is the integer inequality $104976 > 96875$. □

Direct integer calculations give

$$\|p_*\|^2 = 104, \qquad p_*^T D_1 p_* = 28, \qquad \|p_*/2\|^2 = 26.$$

Consequently $H(0) = 52$, $\dot{H}(0) = 24$, and $J = 26T = 13/10000$. Define, only on $[0, T]$,

$$F(t) = H(t) - 52 - \frac{9}{10}\,26t.$$

Then $F(0) = 0$ and $F'(0) = 24 - 117/5 = 3/5 > 0$. Continuity already proves a violation at some positive time. The following bound supplies the explicit time in (4).

Write $y = (p, \Delta)$ and $\mathcal{A} = \left(\begin{smallmatrix} -D_1 & B \\ -B^T & 0 \end{smallmatrix}\right)$. Since $\|B\| \le 2$, $\|D_1\| \le 4$, we have $\|\mathcal{A}\| \le 6$. Also $\|y(0)\| < 11$ and $\|(p_*/2, 0)\| < 6$. Variation of constants, and $e^u \le (1 - u)^{-1}$ for $0 \le u < 1$, yield

$$\|y(t)\| \le e^{6t}(11 + 6t) \le \frac{553}{47} < 12 \qquad (0 \le t \le T < 1/100).$$

For $0 < t < T$, the force is constant and these estimates give

$$\|\dot{y}\| \le 78, \qquad \|\ddot{y}\| \le 468, \qquad |\dddot{H}| \le 78^2 + 12 \cdot 468 = 11700.$$

Apply these estimates on $[0, T]$. Taylor's integral remainder gives

$$F(T) \ge \frac{3}{5}T - 5850T^2 = \frac{123}{8000000} > 0. \tag{5}$$

By Lemma 1, the actual claimed right-hand side is strictly smaller than $52 + (9/10)J$. Therefore

$$H(T) > 52 + \frac{9}{10}J > 52 + \frac{J}{2\sigma_{\min}(D_1)}.$$

This is a fully analytic counterexample to Theorem 4(iii). The estimates do not depend on numerical integration or floating-point eigenvalues.

3 An exact counterexample with zero initial state

How the numerical witness was found

For $N = 128$, $h = 1/2$, $T = 512$, let

$$\mathcal{A}_x = \begin{pmatrix} 0 & I \\ -K & -D_{1/2} \end{pmatrix}, \quad F_x = \begin{pmatrix} 0 \\ I \end{pmatrix}, \quad C_x = \operatorname{diag}(B^T, I), \quad W_T = \int_0^T e^{s\mathcal{A}_x} F_x F_x^T e^{s\mathcal{A}_x^T}\, ds.$$

The largest possible final energy for a unit-L^2 force and zero initial state is $\frac{1}{2}\lambda_{\max}(C_x W_T C_x^T)$; its top eigenvector determines a maximizing force. Numerically this gives energy 43.74150560 against a claimed coefficient 40.78143255. These calculations locate a witness; they are not needed to verify the exact counterexample below.

An exactly specified trajectory and force

Put $m = 2048$, $t_j = j/4$, $M_0 = 2^{40}$ and $Q = 4M_0$. The supplied integer data consist of $X_j, V_j \in \mathbb{Z}^{128}$ for $0 \le j \le m$, with $X_0 = V_0 = 0$. On $[t_j, t_{j+1}]$, set $u = 4(t - t_j)$ and

$$x(t) = Q^{-1} \sum_{k=0}^{3} C_{j,k} u^k, \tag{6}$$

where

$$\begin{aligned} C_{j,0} &= 4X_j, & C_{j,1} &= V_j, \\ C_{j,2} &= 12(X_{j+1} - X_j) - 2V_j - V_{j+1}, & C_{j,3} &= 8(X_j - X_{j+1}) + V_j + V_{j+1}. \end{aligned}$$

These formulas give $x(t_j) = X_j/M_0$ and $\dot{x}(t_j) = V_j/M_0$ on both sides of every knot. Thus x is C^1 and starts from $x(0) = \dot{x}(0) = 0$. *Define*, almost everywhere on $[0, T]$,

$$d = \ddot{x} + D_{1/2}\dot{x} + Kx. \tag{7}$$

After T, set $d = 0$ and continue the homogeneous ODE from the endpoint state. This force is a bounded piecewise polynomial, hence belongs to L^2. Jumps at the finitely many knots cause no difficulty: x and $\dot{x}$ are continuous, so there are no impulsive acceleration terms.

Rounding a numerical trajectory proposed these integer coefficients. They now define a new trajectory exactly, so no interpolation error enters the proof.

Exact input cost and terminal energy

Let $D_2 = 2D_{1/2}$. Its diagonal is $(4, \dots, 4, 3)$, its subdiagonal is -3, and its superdiagonal is -1. The matrix K has diagonal $(2, \dots, 2, 1)$ and adjacent off-diagonals -1. With $C_{j,k} = 0$ for $k > 3$, put

$$G_{j,k} = 2KC_{j,k} + 4(k+1)D_2 C_{j,k+1} + 32(k+1)(k+2)C_{j,k+2}.$$

Then $d(t) = (2Q)^{-1} \sum_{k=0}^{3} G_{j,k} u^k$. Consequently

$$J = \frac{1}{16Q^2} \sum_{j=0}^{m-1} \sum_{k,l=0}^{3} \frac{G_{j,k}^T G_{j,l}}{k + l + 1}, \tag{8}$$

$$H(T) = \frac{\|V_m\|^2 + \left\|B^T X_m\right\|^2}{2M_0^2}. \tag{9}$$

Integer arithmetic and rational summation give

$$J = \frac{16247963052961153388411898917}{162479630156206161080510054 40},$$
$$H(T) = \frac{66100294387661946879318 81}{151115727451828646838272}.$$

In particular, the exact comparisons

$$0 < J < \frac{101}{100}, \qquad H(T) > 43 \tag{10}$$

already suffice for a contradiction.

A certified singular-value comparison

For a positive rational $s = a/b$, form the integer pentadiagonal matrix

$$\mathcal{G}_s = b^2 D_2^T D_2 - 4a^2 I = 4b^2 (D_{1/2}^T D_{1/2} - s^2 I).$$

The verifier performs exact rational LDL^T elimination and reconstructs the factorization. For $s = 3/250$, all 128 pivots are strictly positive. Hence $\mathcal{G}_s \succ 0$ and $\sigma_{\min}(D_{1/2}) > 3/250$. Combining this with (10),

$$\frac{J}{2\sigma_{\min}(D_{1/2})} < \frac{125}{3} J < \frac{505}{12} < 43 < H(T). \tag{11}$$

Since $H(0) = 0$, this explicitly contradicts (3). If an exactly unit-L^2 input is preferred, divide $x, \dot{x}, d$ by $\sqrt{J}$; the inequality is homogeneous, so the contradiction is preserved.

For completeness, the banded elimination uses only the following recurrence. Writing $g_{ij} = (\mathcal{G}_s)_{ij}$, $\ell_{ii} = 1$, and omitting terms with negative indices,

$$\begin{aligned} \ell_{i,i-2} &= g_{i,i-2}/\delta_{i-2}, \\ \ell_{i,i-1} &= (g_{i,i-1} - \ell_{i,i-2}\delta_{i-2}\ell_{i-1,i-2})/\delta_{i-1}, \\ \delta_i &= g_{ii} - \ell_{i,i-1}^2 \delta_{i-1} - \ell_{i,i-2}^2 \delta_{i-2}. \end{aligned}$$

All divisions and signs are evaluated as rational numbers. The certificate lists every pivot, and the checker verifies the identity $\mathcal{G}_s = L\,\mathrm{diag}(\delta_i) L^T$.

Certifying the size of the violation

The same exact elimination gives the sharper bracket

$$\frac{245209}{20000000} < \sigma_{\min}(D_{1/2}) < \frac{24521}{2000000}.$$

The lower matrix has 128 positive pivots. The upper matrix has 127 positive and one negative pivot, so Sylvester's law of inertia proves the upper bound. Using the exact $H(T)$ and J above, these brackets certify

$$7.2581\% < \frac{H(T)}{J/(2\sigma_{\min}(D_{1/2}))} - 1 < 7.25855\%.$$

Thus 7.26% is a justified rounding for the exact polynomial witness, independently of floating-point singular-value computations.

4 Verification, data and scope

The accompanying certificate bundle contains the complete integer trajectory and two verifiers. From the repository root, using only the Python standard library, run

```
python3 certificates/energy-short-time/verify.py
python3 certificates/energy-zero-state/verify.py
```

The first checks the finite integer identities and rational norm/Taylor bounds in §2. The second checks the data, zero initial conditions, knot matching, exact polynomial integrals, every factorization pivot, and the strict energy comparisons in §3. Neither requires NumPy, SciPy, an ODE solver, a matrix exponential or a floating-point eigensolver. Decimal summaries are not used in any proof decision. An optional proposal script explains the numerical origin of the data and is never imported by either checker.

The integer data format is one ASCII JSON header followed by 2049 rows of decimal integers, in the order

$$X_{j,1}, \dots, X_{j,128}, V_{j,1}, \dots, V_{j,128}.$$

The compressed file is `trajectory.txt.gz`; its SHA-256 is

`b2411735af07bf15f213da435844786c417ae3d180760e8948421b304fe2c97f.`

The hash identifies the finite witness; correctness follows from the exact checks, not from a claim that the proposed data are accurate samples. The five-follower argument is entirely analytic. The zero-state argument is a computer-assisted proof by exact rational arithmetic, with its data and checking algorithm supplied. It is not a formalization in a proof assistant.

Smooth compactly supported disturbances also exist: the terminal state depends continuously on d in $L^2(0,T)$ through the bounded map $d \mapsto \int_0^T e^{(T-s)\mathcal{A}_x} F_x d(s)\, ds$. Thus the strict violation persists under sufficiently small smooth L^2 approximations. The explicit certificates use the polynomial forces above.

These examples refute the stated finite coefficient in Theorem 4(iii). They do not refute the separate asymptotic stability and passivity assertions in parts (i)–(ii), or the singular-value estimates of Lemma 6. For a positive definite symmetric part $S = (D_h + D_h^T)/2$, a valid bound follows directly from the energy identity and Young's inequality:

$$H(t) \le H(0) + \frac{1}{4\lambda_{\min}(S)} \int_0^t \|d(s)\|^2 \, ds.$$

Indeed, $-\lambda \|p\|^2 + p^T d \le \|d\|^2 / (4\lambda)$. This supplies a valid replacement estimate; it does not establish the paper's asserted scaling in N for the sharper, system-dependent gain.

References

[1] I. Herman, S. Knorn and A. Ahlén, *Disturbance scaling in bidirectional vehicle platoons with different asymmetry in position and velocity coupling*, Automatica **82** (2017), 13–20. doi:10.1016/j.automatica.2017.04.010. Theorem 4 and equations (3)–(9) are on printed p. 15 in the supplied published PDF; the corresponding author manuscript is dated 15 March 2017.